RAAHAVI KRISHNAKUMAR

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SUMMARY

A dedicated, conscientious, and energetic Data Science undergraduate at SLIIT, with hands-on experience in machine learning, full-stack web development, and mobile application development. Working across multiple academic projects has shaped me into a mature team player who adapts quickly to new technologies and challenges. Proficient in Python, Java, JavaScript, SQL, and the MERN stack, with practical exposure to ML models including CNN, ARIMA, XGBoost, and Random Forest. I can work effectively both independently and in a team, perform well under pressure, and consistently meet deadlines. Seeking an internship or collaborative opportunity to contribute and grow in a data-driven environment.

EDUCATION

BSc (Hons) in Information Technology – Specialising in Data Science — July 2024 – Present
Sri Lanka Institute of Information Technology (SLIIT) | *Expected July 2028*

GCE Advance Level — 2019
J/Uduppiddy girls' College

GCE Odinary Level — 2022
J/Uduppiddy girls' College

PROJECTS

PharmaAI – Medicine Stock Prediction & Inventory Management System — 2026

Group Project | *IT2021 AI & Machine Learning Module (Y2S2)* | *SLIIT* |
github.com/RaahaviKrishanakumar/PharmaAI

- Built a full-stack MERN web application (React.js, Node.js, Express.js, MongoDB) to manage pharmacy inventory, reduce wastage, and enable data-driven stock decisions.
- Led the Sales & Billing Management module: implemented POS transaction flow, billing logic, invoice functionality, and automatic stock deduction after each sale.
- Trained and compared multiple AI/ML models — XGBoost, Random Forest, ARIMA, and ANN — for medicine demand forecasting; **ARIMA selected as the best-performing model**.
- Implemented JWT authentication, bcrypt password hashing, low-stock/expiry alerts, and duplicate medicine detection.

Technologies: MERNStack (React.js, Node.js, Express.js, MongoDB), XGBoost, Random Forest, ARIMA, ANN, JWT, bcrypt

MediCore – Pharmacy Management Mobile App — 2026

Group Project|SE2020Web&MobileTechnologyModule(Y2S2) | *SLIIT* |
github.com/RaahaviKrishanakumar/MediCore

- Developed a full-stack mobile pharmacy management app (Expo React Native frontend, Node.js/Express backend, MongoDB Atlas) covering medicine management, inventory tracking, sales, suppliers, and analytics.
- Contributed the Sales & Billing Management module: built POS flow, cart handling, checkout, sales records, invoice details, and refund functionality.
- Integrated REST APIs with JWT + role-based access control; deployed backend on Render.

Technologies: Expo React Native, Node.js, Express.js, MongoDBAtlas, JWT, REST API, Render

Mental Health in Tech Predictor — 2026

Individual Data Science Project | *https://github.com/RaahaviKrishanakumar/Mental-Health-Predictor* |
Live Demo: https://mental-health-predictor-mg4gnsqdxvhjvww74aywqz.streamlit.app

- Built an end-to-end individual ML pipeline to predict mental health treatment-seeking likelihood among tech workers using the OSMI Mental Health in Tech Survey dataset.
- Performed data cleaning, EDA, and feature encoding across structured Jupyter notebooks covering data understanding, preprocessing, feature engineering, model building, and explainability.
- Trained and compared Logistic Regression, Random Forest, and XGBoost; Random Forest selected as the final model based on F1 Score.
- Applied SHAP to identify key workplace risk factors influencing mental health treatment decisions.
- Deployed an interactive Streamlit web app with real-time risk prediction, data insights, and responsible AI disclaimers.

Technologies: Python, Random Forest, Logistic Regression, XGBoost, SHAP, scikit-learn, Streamlit, Joblib, Pandas, NumPy, Matplotlib, Seaborn

Paddy Disease Classification System – AI & Machine Learning Module (Y2S1) October 2025

Group Project | *SLIIT Faculty of Computing*

- Applied and evaluated multiple ML models (EfficientNetB0, CNN, MobileNetV2, K-Means) to classify 11 paddy disease categories from image data.
- Best-performing CNN model achieved **94.15% accuracy** and a weighted F1-score of 0.9416 across a 1,914-image test dataset.
- Compared precision, recall, and F1-score metrics across all team models to identify the optimal approach for real-world agricultural disease detection.

Technologies: Python, CNN, EfficientNetB0, MobileNetV2, K-Means Clustering, scikit-learn, Image Classification

Web-Based Voting System for Award Nomination – Software Engineering Module (Y2S1) 2025

Group Project |*SLIIT* | *github.com/RaahaviKrishanakumar/Web-Based-Voting-System*

- Designed and implementedthe Financial Managementmoduleforcosttrackingand financial data processing within a multi-user secure voting platform.
- Ensured data consistency and accurate calculations across modular system components built with Java Servlets and JSP.

Technologies: Java, JSP, Servlets, HTML, CSS, MySQL, Apache Tomcat, GitHub

Vehicle Rental System – Object-Oriented Programming Module (Y1S2) 2025

Group Project | *SLIIT* | *github.com/RaahaviKrishanakumar/SLIIT-Vehicle-Rental-System*

- Developeda full-stackweb-basedvehiclerental management system applyingcore OOP principles: encapsulation, inheritance, and polymorphism.
- Builtfront-end withHTML, CSS, andJavaScript; back-endwith Java JSP/Servlets.

Technologies: Java, JSP, Servlets, HTML, CSS, JavaScript, GitHub

Greenhouse Automation System – Fundamentals of Computing Module (Y1S1) 2024

Group Project | *SLIIT*

- Developed a conceptual automationsystemformonitoringand controlling air temperature and soil moisture in a greenhouse environment.

CERTIFICATIONS

AI/ML Engineer–Stage1 2026

AI/ML Engineer–Stage2

Centre for Open & Distance Education, Faculty of Computing, SLIIT

Python for Beginners 2026

Python Programming

Centre for Open & Distance Learning (CODL), University of Moratuwa

SKILLS

Programming Languages: Python, Java, JavaScript, SQL

Web Technologies: React.js, Node.js, Express.js, HTML, CSS, JSP, Servlets, REST APIs, Apache Tomcat

Mobile Development: Expo React Native

Machine Learning / AI: CNN, EfficientNetB0, MobileNetV2, K-Means, XGBoost, Random Forest, ARIMA, ANN, scikit-learn, Logistic Regression, Streamlit, SHAP, Joblib

Data & Visualisation: SQL, Data Analysis, Confusion Matrix, Classification Reporting

Databases: MongoDB, MySQL

Tools & Platforms: GitHub, VS Code, IntelliJ IDEA, JWT, bcrypt, Render, Postman, Streamlit

Soft Skills: Team Collaboration, Analytical Thinking, Problem Solving, Attention to Detail

ACHIEVEMENTS & HIGHLIGHTS

- Contributed to a high-accuracy (94.15%) paddy disease classification ML model as part of a 6-member academic research team.
- Successfully delivered five full-stack software projects across consecutive semesters, including MERN stack web apps, a React Native mobile app, and Java web applications.
- Earned AI/ML Engineer Stage1 & Stage2 certification from SLIIT CODL and two Python certifications from the University of Moratuwa.
- Independently built and deployed an explainable ML app using SHAP and Streamlit to predict mental health treatment likelihood among tech workers, applying responsible AI principles.
- Open to internships, mentorship, and collaborative projects in Data Science, Machine Learning, Full-Stack Development, and Mobile App Development.